

BDBP-307 :- Data Analysis in Genomics , Transcriptome , and Proteomics Laboratory ..

Assignment :- 1

1. Load the dataset sample_data.csv into R

```
list.files()
```

```
'sample_data' · 'sample_data.csv'
```

```
cat(readLines("sample_data.csv"), sep= "\n")
```

```
StudentID, Age, Score, Category  
S01, 21, 45, Biology  
S02, 23, 78, Bioinformatics  
S03, 20, 55, Genomics  
S04, 25, 88, Bioinformatics  
S05, 22, 30, Biology  
S06, 24, 92, Genomics  
S07, 21, 65, Bioinformatics  
S08, 26, 40, Biology  
S09, 23, 72, Genomics  
S10, 22, 81, Bioinformatics
```

2. Store this dataset in a variable as a dataframe

```
my_data = as.data.frame(read.csv("sample_data.csv"))  
my_data
```

A data.frame: 10 × 4

StudentID	Age	Score	Category
<chr>	<int>	<int>	<chr>
S01	21	45	Biology
S02	23	78	Bioinformatics
S03	20	55	Genomics
S04	25	88	Bioinformatics
S05	22	30	Biology
S06	24	92	Genomics
S07	21	65	Bioinformatics
S08	26	40	Biology
S09	23	72	Genomics
S10	22	81	Bioinformatics

3. Write a command to display only the first 6 rows of dataframe to check its column structure.

```
head (my_data , 6)
```

A data.frame: 6 × 4

	StudentID	Age	Score	Category
	<chr>	<int>	<int>	<chr>
1	S01	21	45	Biology
2	S02	23	78	Bioinformatics
3	S03	20	55	Genomics
4	S04	25	88	Bioinformatics
5	S05	22	30	Biology
6	S06	24	92	Genomics

4. Write a second command to view a statistical summary (min, max, median, etc.) for all the variables in the dataset.

```
summary(my_data)
```

StudentID	Age	Score	Category
Length :10	Min. :20.00	Min. :30.00	Length :10
N.unique :10	1st Qu.:21.25	1st Qu.:47.50	N.unique : 3
N.blank : 0	Median :22.50	Median :68.50	N.blank : 0
Min.nchar: 3	Mean :22.70	Mean :64.60	Min.nchar: 7
Max.nchar: 3	3rd Qu.:23.75	3rd Qu.:80.25	Max.nchar:14
	Max. :26.00	Max. :92.00	

5. Write the code to calculate the average (mean) value of the score column.

```
Average_score = mean(my_data$Score)
```

```
[1] 64.6
```

6. Print this calculated value to the console.

```
print(Average_score)
```

7. Write a subsetting or filtering command only contains rows where the Score is strictly greater than 50 and save as new dataframe filter_df

```
filter_df <- my_data[my_data$Score > 50, ]
filter_df
```

A data.frame: 7 × 4

	StudentID	Age	Score	Category
	<chr>	<int>	<int>	<chr>
2	S02	23	78	Bioinformatics
3	S03	20	55	Genomics
4	S04	25	88	Bioinformatics
6	S06	24	92	Genomics
7	S07	21	65	Bioinformatics
9	S09	23	72	Genomics
10	S10	22	81	Bioinformatics

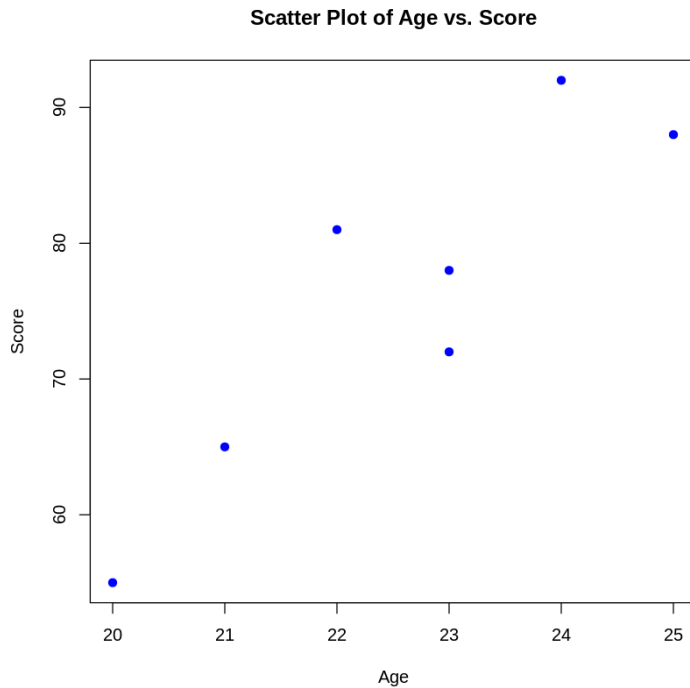
```
filter_df <- subset(my_data, Score > 50)
filter_df
```

A data.frame: 7 × 4

	StudentID	Age	Score	Category
	<chr>	<int>	<int>	<chr>
2	S02	23	78	Bioinformatics
3	S03	20	55	Genomics
4	S04	25	88	Bioinformatics
6	S06	24	92	Genomics
7	S07	21	65	Bioinformatics
9	S09	23	72	Genomics
10	S10	22	81	Bioinformatics

8. plot data as a scatter plot (x-axis to represent Age and the y-axis to rep

```
plot(x = filter_df$Age,
     y = filter_df$Score,
     xlab = "Age",
     ylab = "Score",
     main = "Scatter Plot of Age vs. Score",
     pch = 19,
     col = "blue")
```



✓ Assignment 2 : Sales_data.txt

1. Load the tab-separated dataset sales_data.txt into R.

```
list.files()
```

```
'sales_data.txt' · 'sample_data' · 'sample_data.csv'
```

```
cat(readLines("sales_data.txt"), sep = "\n")
```

OrderID	Category	Quantity	Unit_Price
0001	Sequencing_Consumables	5	120
0002	Bioinformatics_Software	2	450
0003	Lab_Hardware	1	890
0004	Bioinformatics_Software	3	300
0005	Sequencing_Consumables	10	80
0006	Lab_Hardware	2	950
0007	Bioinformatics_Software	1	600
0008	Sequencing_Consumables	4	150

```
sales = read.delim("sales_data.txt", sep="\t")
print(sales)
```

A data.frame: 8 × 4

OrderID	Category	Quantity	Unit_Price
<chr>	<chr>	<int>	<int>
O001	Sequencing_Consumables	5	120
O002	Bioinformatics_Software	2	450
O003	Lab_Hardware	1	890
O004	Bioinformatics_Software	3	300
O005	Sequencing_Consumables	10	80
O006	Lab_Hardware	2	950
O007	Bioinformatics_Software	1	600
O008	Sequencing_Consumables	4	150

3. Modify your sales data frame to include a new column called Total_Price.

```
sales$Total_Price <- sales$Quantity * sales$Unit_Price
```

A data.frame: 8 × 5

OrderID	Category	Quantity	Unit_Price	Total_Price
<chr>	<chr>	<int>	<int>	<int>
O001	Sequencing_Consumables	5	120	600
O002	Bioinformatics_Software	2	450	900
O003	Lab_Hardware	1	890	890
O004	Bioinformatics_Software	3	300	900
O005	Sequencing_Consumables	10	80	800
O006	Lab_Hardware	2	950	1900
O007	Bioinformatics_Software	1	600	600
O008	Sequencing_Consumables	4	150	600

4. Calculate the values for this column by multiplying the Quantity column by the Unit_Price column for every row.

```
print(sales)
```

5. Create a new summary table or data frame called category_summary.

```
category_summary <- aggregate(Total_Price ~ Category, data = sales, FUN = sum)

# Rename the column for clarity
names(category_summary)[2] <- "Total_Revenue"

# View the result
category_summary
```

A data.frame: 3 × 2

Category	Total_Revenue
<chr>	<int>
Bioinformatics_Software	2400
Lab_Hardware	2790
Sequencing_Consumables	2000

6. Write a command that groups your dataset by the unique values in the Category column.

```
grouped_sales_list <- split(sales, sales$Category)
grouped_sales_list
```

\$Bioinformatics_Software

A data.frame: 3 × 5

	OrderID	Category	Quantity	Unit_Price	Total_Price
	<chr>	<chr>	<int>	<int>	<int>
2	O002	Bioinformatics_Software	2	450	900
4	O004	Bioinformatics_Software	3	300	900
7	O007	Bioinformatics_Software	1	600	600

\$Lab_Hardware

A data.frame: 2 × 5

	OrderID	Category	Quantity	Unit_Price	Total_Price
	<chr>	<chr>	<int>	<int>	<int>
3	O003	Lab_Hardware	1	890	890
6	O006	Lab_Hardware	2	950	1900

\$Sequencing_Consumables

A data.frame: 3 × 5

	OrderID	Category	Quantity	Unit_Price	Total_Price
	<chr>	<chr>	<int>	<int>	<int>
1	O001	Sequencing_Consumables	5	120	600
5	O005	Sequencing_Consumables	10	80	800
8	O008	Sequencing_Consumables	4	150	600

7. Calculate the total revenue generated for each category by summing up your newly created Total_Price values.

```
# 1. Re-load the dataset from the file
sales <- read.delim("sales_data.txt")

# 2. Re-create the Total_Price column
sales$Total_Price <- sales$Quantity * sales$Unit_Price
```

```
# 3. Calculate total revenue by category (Task 7)
category_summary <- aggregate(Total_Price ~ Category, data = sales, FUN = sum)

# 4. Rename the column to 'Total_Revenue'
names(category_summary)[2] <- "Total_Revenue"

# Display the resulting dataframe
category_summary
```

A data.frame: 3 × 2

Category	Total_Revenue
<chr>	<int>
Bioinformatics_Software	2400
Lab_Hardware	2790
Sequencing_Consumables	2000

8. Print this summary table.

```
# Task 8: Print the summary table
print(category_summary)

# Task 9: Create a bar chart
barplot(height = category_summary$Total_Revenue,
        names.arg = category_summary$Category,
        xlab = "Category",
        ylab = "Total_Revenue",
        main = "Total Revenue by Category",
        col = "steelblue",
        las = 1,           # Makes y-axis labels horizontal
        cex.names = 0.8)  # Slightly shrinks x-axis labels so they don't cut off
```

	Category	Total_Revenue
1	Bioinformatics_Software	2400
2	Lab_Hardware	2790
3	Sequencing_Consumables	2000

